

Felix Dülmer

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EDUCATION

01/2023 – Present	Ph.D. Computer Science Focus: Robotic Ultrasound and Differentiable Ultrasound Simulation Advisor: Prof. Nassir Navab	<i>Technical University of Munich, Germany</i>
10/2018 – 09/2022	M.Sc. Mechatronics and Robotics Focus: Robot-Assisted Medical Systems Grade: 1.5	<i>Technical University of Munich, Germany</i>
07/2019 – 12/2019	Exchange Semester Postgraduate Focus: Deep Learning and Big Data Analysis	<i>Universidad de los Andes, Colombia</i>
04/2019 – 06/2019	Autonomous Driving Project at Cognitive Systems Lab Test of a Tactile Steering Wheel in Autonomous Driving Grade: 1.0	<i>University of Tsukuba, Japan</i>
10/2014 – 10/2018	B.Sc. Mechanical Engineering Focus: Autonomous Driving Grade: 2.5	<i>Technical University of Munich, Germany</i>

EXPERIENCE

01/2023 - Present	PhD Candidate in Computer Science	<i>Technical University of Munich, Germany</i>
Robotic Ultrasound & Automation [2, 7, 9]: Developed ML methods for quality-aware robotic ultrasound, including Doppler-based quality estimation for robust acquisition [9]. Contributed to a field-level perspective on challenges and opportunities in ML in robotic ultrasound, with links to translational robot-assisted vascular surgery [7, 2]. Physics-Based & Differentiable Ultrasound Modeling [1, 3–6]: Built physics-informed, differentiable ultrasound image-formation and simulation methods, spanning ray tracing and learned 3D representations [3–5]. Advanced model-based enhancement and novel-view synthesis via deconvolution and Gaussian-based ray casting [6, 1]. Leadership & Supervision: Supervised 10+ students and coordinated research activities across chairs. Contributed to grant writing and proposal development.		
03/2026 - 05/2026	Visiting Researcher	<i>Stanford University, Ultrasound Imaging and Instrumentation Lab</i>
Developed differentiable ultrasound models for bent-ray propagation and continuous speed-of-sound estimation under the supervision of Prof. Jeremy Dahl.		
04/2021 – 09/2022	Internship/Working Student as Software Developer	<i>Workerbase GmbH, Munich, Germany</i>
Developed and maintained two cross-platform industrial mobile apps deployed to thousands of factory workers, using Android (Kotlin/Java), iOS (Swift), and Flutter (Dart). Applied clean architecture, design patterns, and test-driven development to ensure scalable, reliable delivery.		
09/2017 – 08/2020	Research Assistant	<i>Technical University of Munich, Germany</i>
Maintained and extended a driving simulator for automated driving research, and onboarded/trained students on its use and related software platforms. Contributed to prototyping components supporting higher levels of vehicle autonomy.		

SERVICES AND AWARDS

- Reviewer: IPCAI; MICCAI; ECCV; *IEEE Transactions on Automation Science and Engineering (T-ASE)*
- Scholarship: MCML AI-X Change (2025/2026)
- Reviewer Award: Honorable Mention MICCAI 2025
- Best Paper Runner-Up Award: ASMUS Workshop @ MICCAI 2024
- Hamlyn Winter School: Surgical Imaging and Vision Attendance and Runner-Up Award (2022)

SKILLS

Programming Languages & Frameworks:	Python; PyTorch; Dr.Jit; Mitsuba; C++; Java; Swift; Dart (Flutter); MATLAB
Robotics & Middleware:	ROS (catkin/rosbag/TF); RViz; URDF/Xacro; Gazebo
Systems / IT:	Unix/Linux; Slurm; Git; Web programming
Tools / IDEs:	VS Code; GitHub Copilot; Cursor (AI-assisted coding)
Languages:	German (native); English (fluent); Spanish (proficient); French (basic)

PUBLICATIONS

1. **Felix Dülmer**; Jakob Klaushofer; Magdalena Wysocki; Nassir Navab; Mohammad Farid Azampour. UltraG-Ray: Physics-Based Gaussian Ray Casting for Novel Ultrasound View Synthesis. 2026, *Medical Imaging with Deep Learning*.
2. **Felix Dülmer**; Mario Reidl; Nikolina-Alexia Fasoula; Nikoletta Katsouli; Dianyue Huang; Yuan Bi; Raphael Salz; Rhiannon Lees; Zhongliang Jiang; Christoph Knappich; *et al.* Translational Research and Clinical Trends in Robot-Assisted Vascular Surgery: A Narrative Overview. 2026, *Handbook of Robotic and Image-Guided Surgery* (Elsevier).
3. **Felix Dülmer**; Mohammad Farid Azampour; Nassir Navab. UltraScatter: Ray-Based Simulation of Ultrasound Scattering. 2025, *IEEE International Ultrasonics Symposium (IUS)*.
4. **Felix Dülmer**; Mohammad Farid Azampour; Magdalena Wysocki; Nassir Navab. UltraRay: Introducing Full-Path Ray Tracing in Physics-Based Ultrasound Simulation. 2025, *MICCAI (International Conference on Medical Image Computing and Computer-Assisted Intervention)*.
5. Magdalena Wysocki; **Felix Dülmer**; Ananya Bal; Nassir Navab; Mohammad Farid Azampour. UltrON: Ultrasound Occupancy Networks. 2025, *MICCAI (International Conference on Medical Image Computing and Computer-Assisted Intervention)*.
6. **Felix Dülmer**; Walter Simson; Mohammad Farid Azampour; Magdalena Wysocki; Angelos Karlas; Nassir Navab. Phocus: Physics-based deconvolution for ultrasound resolution enhancement. 2024, *International Workshop on Advances in Simplifying Medical Ultrasound*.
7. Yuan Bi; Zhongliang Jiang; **Felix Dülmer**; Dianyue Huang; Nassir Navab. Machine learning in robotic ultrasound imaging: Challenges and perspectives. 2024, *Annual Review of Control, Robotics, and Autonomous Systems*.
8. Angelos Karlas; Nikolina-Alexia Fasoula; Michael Kallmayer; Christoph Schäffer; Georgios Angelis; Nikoletta Katsouli; Mario Reidl; Felix Dülmer; Kenana Al Adem; Leontios Hadjileontiadis; *et al.* **Optoacoustic biomarkers of lipids, hemorrhage and inflammation in carotid atherosclerosis**. 2023, *Frontiers in Cardiovascular Medicine*.
9. Zhongliang Jiang*; **Felix Dülmer***; Nassir Navab. Dopus-net: Quality-aware robotic ultrasound imaging based on doppler signal. 2023, *IEEE Transactions on Automation Science and Engineering*.

* shared first authorship